

# Fair and Constraint-Aware Group Formation

An interpretable decision-support prototype

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## Abstract

We design and implement a decision-support system that forms fair and balanced groups in collaborative settings such as Erasmus+ youth exchanges. The task is modelled as a constraint-based optimization problem: participants are partitioned into fixed-size groups (hard constraints) so as to minimize a weighted combination of soft penalties capturing skill balance, diversity, and preference satisfaction. We implement and compare four methods: a random baseline, swap-based local search, simulated annealing, and an integer-linear-programming (ILP) formulation, then we evaluate them with fairness metrics that are deliberately independent of the optimization objective, analyse robustness under participant addition/removal, and provide faithful per-assignment explanations. On a synthetic cohort of 60 participants, the optimized methods roughly halve the cost of the random baseline; warm-started re-solving keeps assignments stable under perturbation; and a Pareto analysis makes the skill-versus-preference trade-off explicit. We close with an ethical discussion of fairness, inclusion, and transparency, framing the tool as support for (not a replacement of) human judgement.

## 1 Introduction

The composition of collaborative groups has a significant influence on participation, inclusion, communication, and learning outcomes. In programmes such as Erasmus+ youth exchanges, organisers often need to divide participants with different backgrounds, skills, and preferences into balanced teams. Achieving this manually can be time-consuming and subjective, particularly when several competing objectives must be considered simultaneously. Simple approaches such as random assignment are easy to apply but frequently produce unbalanced or unfair group compositions.

From a computational perspective, this task can be formulated as a constrained optimisation problem closely related to the *Maximally Diverse Grouping Problem* [1] and other team-formation problems [2]. However, unlike purely mathematical optimisation problems, group formation also raises important ethical questions. Fairness cannot be reduced to a single numerical objective, and different stakeholders may reasonably prioritise different aspects of fairness, such as diversity, equal skill distribution, or participant preferences. Consequently, optimisation should be viewed as a tool for supporting human decision-making rather than replacing it.

This project investigates how group formation can be partially automated while remaining fair, robust, and transparent. Specifically, it aims to answer the following research question:

*How can optimisation techniques be used to generate fair and explainable group assignments while balancing multiple competing objectives?*

The main contributions of this work are:

- a formal constraint-based model for fair group formation;
- four optimisation methods spanning heuristic and exact approaches;
- an evaluation framework based on fairness metrics independent of the optimisation objective;

- a robustness analysis under participant addition and removal;
- a lightweight explainability layer for assignment decisions; and
- an ethical discussion of fairness, transparency, and human oversight.

The resulting system is implemented as a Python command-line application and serves as a prototype demonstrating how optimisation techniques can support fairer and more transparent group formation.

## 2 Problem Modelling

The goal of the system is to automatically divide participants into balanced groups while satisfying a number of practical requirements. Some requirements are mandatory, such as ensuring that every participant belongs to exactly one group and that all groups contain the correct number of members. Others represent desirable properties, such as balancing participant skills, promoting diversity, and respecting personal preferences whenever possible.

Since these objectives may conflict with one another, the problem is formulated as a constraint-based optimisation task. Our system attempts to find the assignment that provides the best overall balance between the different objectives while always satisfying the mandatory constraints.

Let  $P = \{1, \dots, n\}$  denote the set of participants and  $\mathcal{G} = \{g_1, \dots, g_K\}$  the set of groups. An assignment is represented by the function

$$\pi : P \rightarrow \mathcal{G},$$

**Hard constraints.** Hard constraints define the conditions that every valid solution must satisfy. These constraints cannot be violated regardless of how much they might improve the optimisation objective. Every participant belongs to exactly one group, and each group  $g$  has a fixed capacity  $c_g$ :

$$\sum_{g \in \mathcal{G}} \mathbf{1}[\pi(i) = g] = 1 \quad \forall i \in P, \quad \sum_{i \in P} \mathbf{1}[\pi(i) = g] = c_g \quad \forall g \in \mathcal{G}. \quad (1)$$

Capacities form a balanced partition of  $n$  (sizes differ by at most one). Together, these constraints ensure that every participant is assigned to exactly one group and that all groups have the intended size. Any assignment satisfying these conditions is considered feasible and can be evaluated by the optimisation algorithms.

**Soft constraints.** Soft constraints represent desirable properties rather than strict requirements. Because it is often impossible to satisfy all of them simultaneously, they are converted into numerical penalty values. Lower penalties correspond to fairer or more desirable group assignments.

Each participant  $i$  has a skill vector  $s_i \in [0, 1]^d$ . Let  $m_g = \frac{1}{|g|} \sum_{i \in g} s_i$  be the per-group skill mean. The *skill-balance* penalty is the across-group variance of these means, averaged over skills and normalized by the maximal variance 0.25 of values in  $[0, 1]$ :

$$\Phi_{\text{skill}}(\pi) = \frac{1}{d} \sum_{j=1}^d \frac{\text{Var}_{g \in \mathcal{G}}(m_{g,j})}{0.25}. \quad (2)$$

For each categorical attribute  $a$  (e.g. nationality, gender) with global distribution  $q_a$  and per-group distribution  $p_{g,a}$ , the *diversity* penalty is the mean total variation distance to the global mix:

$$\Phi_{\text{div}}(\pi) = \frac{1}{|A|K} \sum_{a \in A} \sum_{g \in \mathcal{G}} \frac{1}{2} \sum_c |p_{g,a}(c) - q_a(c)|. \quad (3)$$

Let  $R$  be the set of directed preference relations: a *like* ( $i \rightarrow j$ ) is satisfied when  $\pi(i) = \pi(j)$ , a *dislike* when  $\pi(i) \neq \pi(j)$ . The *preference* penalty is the unsatisfied fraction:

$$\Phi_{\text{pref}}(\pi) = 1 - \frac{|\{r \in R : r \text{ satisfied}\}|}{|R|}. \quad (4)$$

**Objective.** The optimizers minimize the weighted cost

$$C(\pi) = w_{\text{skill}} \Phi_{\text{skill}}(\pi) + w_{\text{div}} \Phi_{\text{div}}(\pi) + w_{\text{pref}} \Phi_{\text{pref}}(\pi), \quad (5)$$

with weights  $w$  encoding the organizer’s priorities. Unless stated otherwise we use  $w_{\text{skill}} = 1$ ,  $w_{\text{div}} = 1$ ,  $w_{\text{pref}} = 0.5$ .

The objective function combines the individual penalties into a single optimisation score. The weighting parameters determine the relative importance of each fairness criterion and allow organisers to prioritise different aspects of group formation. For example, an educational programme may place greater emphasis on balancing participant skills, whereas a cultural exchange may prioritise diversity or social preferences.

The optimisation algorithms aim to minimise this overall cost while always satisfying the hard constraints. Consequently, the final solution represents a compromise between several competing objectives rather than the perfect optimisation of any single criterion.

### 3 Fairness Metrics

Optimising the objective function alone does not necessarily guarantee that the resulting groups are fair. An algorithm may achieve a low optimisation cost simply because it has been designed to minimise that particular objective, without performing equally well when viewed from other perspectives. For this reason, the quality of the generated assignments is evaluated using a separate set of fairness metrics that are computed independently of the optimisation objective presented in Equation 5.

Using independent evaluation metrics provides a more objective assessment of solution quality and allows different optimisation methods to be compared fairly. Rather than measuring only how well an algorithm minimises its own objective, these metrics assess whether the final group assignments satisfy broader fairness goals. This distinction is particularly important in ethical AI applications, where fairness is a multi-dimensional concept and cannot be fully represented by a single numerical score.

The evaluation considers four complementary metrics, each capturing a different aspect of fair group formation. One of these metrics, experience imbalance, is intentionally evaluated separately rather than included in the optimisation objective. We chose to optimise only skill balance, diversity, and participant preferences, while using experience balance as an independent measure of solution quality. This allows us to evaluate whether the optimisation also improves aspects of fairness that were not explicitly included in the objective function.

**Skill imbalance.** One of the primary goals of the system is to distribute participant skills as evenly as possible across all groups. Skill imbalance is measured as the variance of the average skill level of each group. A lower variance indicates that groups have similar overall abilities, reducing the likelihood that one group is significantly stronger or weaker than the others. Balanced skill distribution helps ensure that participants have comparable opportunities to contribute and learn during collaborative activities.

**Experience imbalance.** In addition to technical skills, participants may have different levels of previous experience. If highly experienced participants are concentrated within a small number of groups, other groups may have fewer opportunities for peer learning and knowledge sharing.

The experience imbalance metric measures the variance of average experience levels across groups. Lower values indicate that experienced participants are distributed more evenly, creating a more balanced collaborative environment.

**Diversity imbalance.** An important objective of programmes such as Erasmus+ is to encourage interaction between participants from different backgrounds. Diversity imbalance measures how closely the demographic composition of each group reflects the overall composition of the participant population. This is calculated using the total variation distance between each group’s distribution and the global distribution of categorical attributes such as nationality or gender. Smaller values indicate that the groups are more representative of the overall cohort, promoting inclusion and intercultural collaboration.

**Preference satisfaction.** Participants may also express preferences regarding people they would like to work with or prefer to avoid. Rather than measuring the number of violated preferences, this metric reports the proportion of preferences that are successfully respected in the final assignment. Higher values therefore indicate greater participant satisfaction while still allowing the optimisation process to balance individual wishes against other fairness objectives.

Taken together, these metrics provide a broader evaluation of fairness than the optimisation objective alone. They acknowledge that fairness is not a single measurable property but instead involves balancing several competing considerations, including equal opportunities, diversity, and participant satisfaction. Improvements in one metric may sometimes require compromises in another, making trade-offs an unavoidable part of the optimisation process.

Finally, these metrics are intended to support human decision-making rather than replace it. They provide quantitative evidence about the quality of different group assignments while recognising that organisers may possess additional contextual information that cannot easily be represented mathematically, such as interpersonal dynamics or accessibility requirements. Consequently, the optimisation results should be viewed as informed recommendations that assist, rather than replace, human judgement.

## 4 Algorithms

To investigate different approaches to fair group formation, we implemented four solution methods. These range from a simple random baseline to heuristic optimisation algorithms and an exact mathematical optimisation model [3, 4, 5]. By comparing these methods, we evaluate performance of different optimisation strategies.

**Random baseline.** We first implemented a random baseline that generates a valid assignment while respecting all hard constraints, such as group size. Although this method does not optimise any fairness objective, it provides an important reference point for evaluating the performance of the optimisation algorithms. Any improvements achieved by the other methods can therefore be measured against a simple random assignment.

**Local Search (Hill Climbing).** For our first optimisation approach, we implemented a Local Search algorithm. We begin with a random feasible assignment and repeatedly attempt to improve it by swapping participants between different groups. A proposed swap is accepted only if it reduces the overall objective function defined in Equation 5. This process continues until no further improvements can be found or a predefined iteration limit is reached.

To make the algorithm more efficient, we implemented incremental cost evaluation. Instead of recalculating the entire objective after every swap, we update only the groups and participant relationships affected by that swap. This significantly reduces computation time while producing

the same optimisation result. One limitation of Local Search is that it always accepts only improving moves, meaning it can become trapped in locally optimal solutions.

**Simulated Annealing (SA).** We also implemented Simulated Annealing, which extends the Local Search algorithm by occasionally accepting swaps that temporarily worsen the solution. The probability of accepting these moves depends on a temperature parameter that gradually decreases throughout the optimisation process [4]. Early in the search, the algorithm explores a wide range of possible solutions, while later it behaves more like standard Local Search.

Allowing temporary decreases in solution quality helps the algorithm escape local optima and explore a larger portion of the search space. In our implementation, we keep track of the best assignment encountered during the optimisation and return that solution when the search finishes.

**Integer Linear Programming (ILP).** Finally, we implemented an Integer Linear Programming (ILP) formulation using the OR-Tools CP-SAT solver [6]. In this model, binary decision variables determine whether each participant is assigned to a particular group while ensuring that all hard constraints are always satisfied.

Because our original objective function contains non-linear components, particularly the variance used to measure skill balance, it cannot be represented directly within a linear optimisation model. We therefore constructed a linear approximation of the objective. Preference satisfaction is modelled exactly, diversity is represented with minor rounding approximations, and skill balancing is approximated by minimising the deviation of each group’s total skill from its target value rather than directly minimising variance.

The ILP approach allows us to compare the heuristic methods against a more exact optimisation technique. However, exact optimisation becomes increasingly computationally expensive as the problem size grows, making heuristic approaches more practical for larger participant cohorts.

## 5 Experimental Setup

To evaluate the proposed methods, we generated a synthetic dataset consisting of 60 participants. We generated a synthetic dataset because no suitable public dataset containing participant skills, diversity attributes, and interpersonal preferences was available. Using synthetic data allowed us to control the participant characteristics, repeat experiments under identical conditions, and avoid privacy concerns associated with personal information. Although synthetic data cannot perfectly represent real Erasmus+ or other participants, they provide a suitable environment for evaluating the optimisation methods.

For each participant, we generated four skill values, one of five experience levels, two diversity attributes (nationality and gender), and optional preferences indicating people they would like or prefer not to work with. We used the **skewed-skill** scenario, where a relatively small number of participants have high skill levels while the majority have lower skill values. This creates a realistic optimisation problem in which balancing group abilities becomes challenging.

We divided the participants into 10 groups of six members each. To reduce the effect of random initialisation, we executed every optimisation method using five different random seeds. For the ILP solver, we imposed a time limit of 10 seconds to ensure reasonable execution times while still allowing the solver to search for high-quality solutions.

Finally, we automated the complete experimental workflow using the script `scripts/run_analysis.py`. This script regenerates the dataset, executes all optimisation methods, computes the evaluation metrics, and reproduces every figure presented in this report, ensuring that the experiments are fully reproducible.

## 6 Results

### 6.1 Method comparison

Table 1 reports means over 5 seeds; Figure 1 visualizes them. The optimized methods roughly halve the cost of the random baseline. SA achieves the lowest cost (0.342), with local search close behind (0.355) at a fraction of the runtime. ILP attains by far the best *skill* balance (0.0008 vs 0.003–0.007), exactly as its  $L_1$  surrogate intends, but spends that advantage elsewhere: its preference satisfaction is the lowest of the optimized methods, so its overall cost is higher. This is the surrogate-versus-metric tension anticipated in Section 4, and a useful reminder that “optimal” is only optimal *with respect to the encoded objective*.

Table 1: Mean results over 5 seeds ( $n = 60$ , groups of 6). Lower is better except preference satisfaction.

| Method       | Cost   | Skill imb.    | Exp. imb. | Div. imb. | Pref. sat. | Runtime (s) |
|--------------|--------|---------------|-----------|-----------|------------|-------------|
| SA           | 0.3422 | 0.0029        | 0.1878    | 0.0900    | 0.5190     | 0.56        |
| Local search | 0.3551 | 0.0035        | 0.1844    | 0.0958    | 0.5095     | 0.19        |
| ILP          | 0.4215 | <b>0.0008</b> | 0.1500    | 0.1110    | 0.3857     | 10.04       |
| Random       | 0.7084 | 0.0073        | 0.3022    | 0.2830    | 0.2071     | 0.00        |

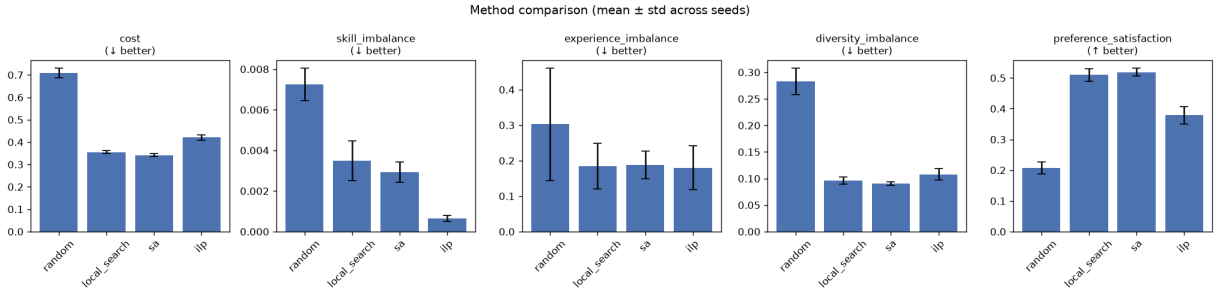


Figure 1: Per-metric comparison (mean  $\pm$  std across seeds). Experience imbalance was not explicitly included in the optimisation objective because we focused on optimising skill balance, diversity, and participant preferences. We evaluate it separately to examine whether the optimisation also improves fairness in aspects that were not directly optimised.

Experience imbalance is not explicitly included in the optimisation objective, yet all optimisation methods still produce noticeably better experience balance than random assignment. Although the algorithms primarily optimise skill balance, diversity, and participant preferences, improving these objectives also leads to more balanced experience levels across the groups.

Overall, we found that Local Search provides the best balance between optimisation quality and computational efficiency. Although Simulated Annealing achieved the lowest optimisation cost, the improvement over Local Search was relatively small while requiring more computation. The ILP approach produced the strongest skill balancing but required substantially longer execution time and optimised a linear approximation of the original objective. For practical group formation tasks, Local Search therefore appears to provide the most suitable compromise.

### 6.2 Convergence and skill distribution

Figure 2 shows cost decreasing monotonically for hill climbing and non-monotonically (with accepted uphill moves) for SA, which reaches a slightly lower final cost. Figure 3 shows the spread of per-group mean skill: tight for the optimized methods (balanced groups) and wide for random.

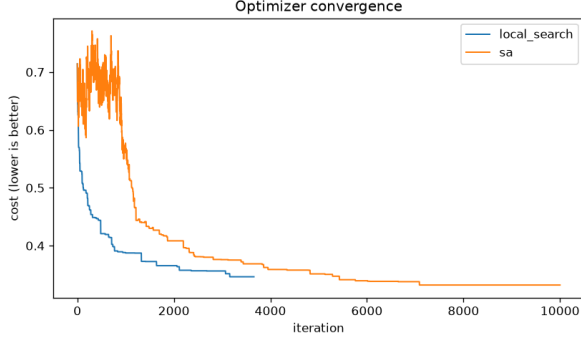


Figure 2: Cost vs iteration.

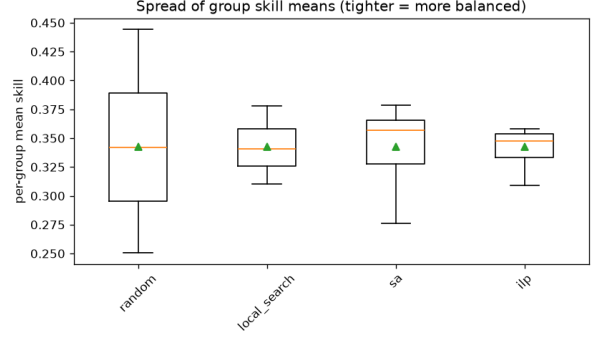


Figure 3: Per-group mean skill spread.

### 6.3 Robustness

We perturb the cohort by removing or adding  $k \in \{1, 3, 6\}$  participants and re-solve with local search, comparing *cold* starts (fresh random initialization) against *warm* starts (re-optimizing from the previous solution, repaired to the new capacities). Stability is measured by the **group-change rate** (fraction of common participants who change group, after aligning arbitrary group labels by maximum overlap) and the label-invariant **pair-preservation** rate (fraction of participant pairs whose same-group status is preserved). Table 2 and Figure 4 summarize the study.

Table 2: Mean stability over 5 trials (local search). Warm-starting drastically reduces churn while fairness barely drifts.

| Perturb. | Start | $k$ | Group-change rate | Pair preservation |
|----------|-------|-----|-------------------|-------------------|
| remove   | cold  | 3   | 0.600             | 0.865             |
| remove   | warm  | 3   | 0.312             | 0.918             |
| add      | cold  | 3   | 0.570             | 0.866             |
| add      | warm  | 3   | 0.107             | 0.965             |

Warm-starting reduces the group-change rate by roughly 2–5 $\times$  relative to cold starts (e.g. 0.107 vs 0.570 for adding three participants) while keeping pair-preservation high. Across all settings the change in skill imbalance after perturbation is on the order of  $10^{-3}$ , i.e. fairness is essentially preserved. For a real organizer, this means a late change to the roster need not reshuffle everyone: warm-started re-solving yields a stable, minimally disruptive update.

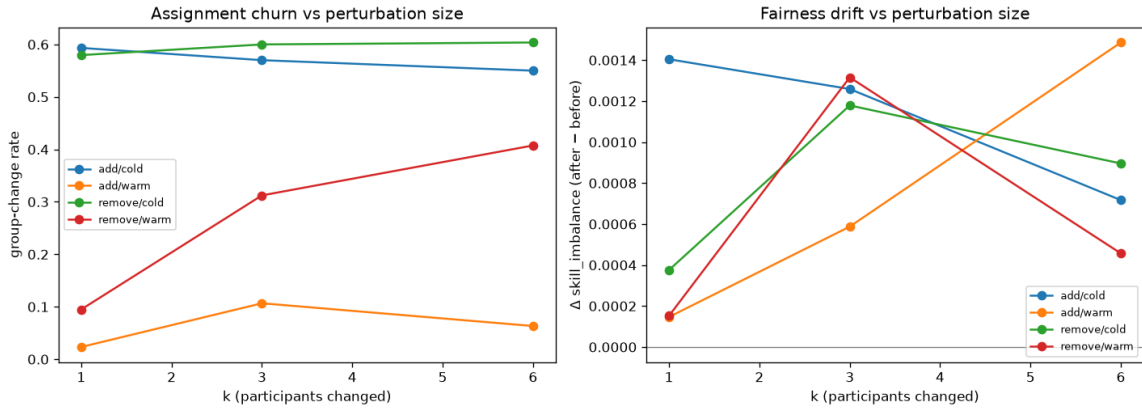


Figure 4: Assignment churn (left) and fairness drift (right) vs perturbation size  $k$ , for add/remove and cold/warm starts.

This is particularly valuable in practical settings such as Erasmus+ exchanges, where participant lists often change shortly before an activity begins and organisers wish to minimise unnecessary disruption.

## 6.4 The fairness trade-off (Pareto front)

Soft objectives conflict: pushing skill balance can cost preference satisfaction. Sweeping the weight  $\alpha$  between skill ( $w_{\text{skill}} = \alpha$ ) and preference ( $w_{\text{pref}} = 1 - \alpha$ ) traces the trade-off in Figure 5. The non-dominated (Pareto-optimal) settings lie at high  $\alpha$ : at  $\alpha = 1$  skill imbalance is minimal (0.00025) but preference satisfaction collapses to 0.27; at  $\alpha = 0.8$  preference satisfaction peaks (0.64) at a small skill cost. Notably, for  $\alpha \leq 0.7$  the solver converges to essentially the same solution — the trade-off only “bites” near the extremes. There is no single fair assignment: the choice along this curve is a *value judgement* that should rest with a human, which the system makes explicit rather than hiding inside a default weight.

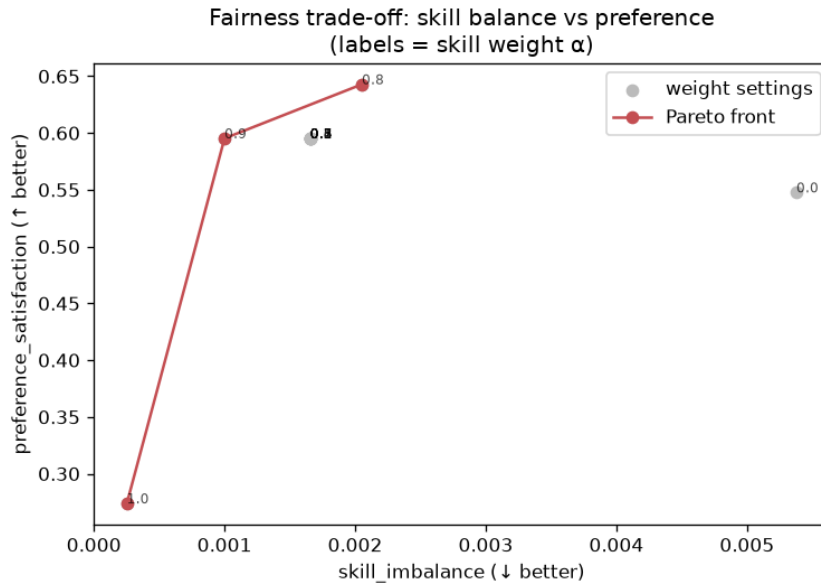


Figure 5: Skill balance vs preference satisfaction as the weight  $\alpha$  varies. Labels show  $\alpha$ ; the red curve is the Pareto front.

## 6.5 Explainability

For each participant, the system generates an explanation directly from the optimisation model [11]. Because group sizes are fixed, the natural counterfactual is a *swap*: we report the single swap that would most reduce cost (using the same incremental evaluator the optimizer uses), or state that the participant is “well-placed” if no swap improves the cost. On the converged Local Search solution for our canonical dataset (seed 42), 97% of participants (58/60) are well-placed; the remaining two differ only at numerical precision. A concrete example:

Participant p000 → group g8

- Placement: well-placed (no beneficial swap found)
- Preferences: 1/2 satisfied --- grouped with liked p058; NOT grouped with liked p039
- Diversity: nationality=PL 17% in group vs 17% overall (balanced); gender=M 50% in group vs 38% overall (over-represented)
- Skill: mean skill 0.21 lowers group mean 0.36 (overall 0.34)

Each factor is derived from the actual soft constraints, not reconstructed post-hoc by a separate explanation model, so the explanation reflects the same criteria used during optimisation.



## 7 Ethical Discussion

Automating group formation directly affects fairness, inclusion, and transparency [7, 10]. Although optimization algorithms can produce balanced group assignments, they cannot determine what is universally "fair". Instead, they optimize the objectives that we choose to include in the model. This makes it important to carefully consider which fairness criteria are represented and to keep a human decision-maker involved in the final decision.

**Fairness is plural and contested.** Our system operationalizes fairness as a combination of skill balance, demographic representativeness, and preference satisfaction. These objectives can conflict (Figure 5) and, in some cases, different fairness criteria cannot all be satisfied simultaneously [9]. Consequently, there is no single objectively "fairest" weighting of the optimization objectives [8]. Instead of hiding these decisions inside the algorithm, we expose the weights as explicit parameters that organisers can adjust according to their priorities. The Pareto analysis further makes these trade-offs visible, allowing users to understand the consequences of different fairness choices.

**Inclusion.** We measure diversity relative to the overall participant population, encouraging minority groups to be distributed across different teams rather than concentrated in only a few. Diversity metric rewards matching the overall population distribution, minority participants tend to be spread across multiple groups. However, this also highlights an important ethical consideration. If a minority group is very small, evenly distributing its members may result in every participant becoming the only representative of that group. Although this improves the numerical diversity metric, it may not always produce the most inclusive outcome. This illustrates that fairness metrics cannot fully capture every aspect of inclusion, and that organisers should retain the ability to review and adjust the generated assignments when appropriate.

Our generator supports a minority-underrepresented scenario, but the quantitative results in this report use the skewed-skill scenario.

**Transparency and contestability.** Every assignment is accompanied by a human-readable explanation together with a counterfactual alternative (Section "Explainability"). This allows organisers to understand why a participant was assigned to a particular group, what factors influenced that decision, and what alternative assignment would most improve the objective function. Such explanations support transparency and make it easier to question or override recommendations when necessary.

We also report stability using label-invariant measures because group labels themselves have no intrinsic meaning. Small design choices such as this help ensure that the reported evaluation reflects genuine assignment changes rather than arbitrary implementation details.

**Decision support, not automation.** Throughout this project, we view the proposed system as a decision-support tool rather than a replacement for human judgement. The optimizer can only improve the objectives that have been explicitly included in the optimization model. Experience balance was not part of our objective function, yet the optimised methods still produced noticeably better experience balance than random assignment. This shows that evaluation metrics can reveal improvements beyond the explicit optimisation target.

From our experiments, we learned that the system only optimises what we ask it to optimise. If we leave out an important fairness criterion, the algorithm cannot consider it. Therefore, we believe the proposed system should be used to support organisers rather than replace them, allowing people to consider factors that cannot easily be included in the optimisation model.

**Limitations.** Our work has several limitations. First, the evaluation is based entirely on synthetic data. Although this allows controlled and reproducible experiments, synthetic participants

cannot fully capture the complexity or historical biases present in real-world datasets. Second, the ILP model optimizes a linear surrogate of the original objective, meaning that it is optimal with respect to an approximation rather than the exact optimisation function. Third, reducing participants to skill vectors, demographic attributes, and preference relationships inevitably simplifies many important human characteristics. Finally, our preference model considers only simple likes and dislikes and does not capture more complex interpersonal relationships or power dynamics between participants.

## 8 Conclusion and Future Work

In this project, we designed and implemented an interpretable, constraint-aware decision-support system for fair group formation. We compared heuristic and exact optimization methods and showed that optimization can reduce the objective cost by approximately half compared with random assignment. We also demonstrated that warm-started re-optimization maintains stable assignments when participant lists change, while the Pareto analysis makes the trade-off between competing fairness objectives explicit.

An important finding of this work is that the evaluation metrics are intentionally broader than the optimization objective. This allows us to identify aspects of fairness that are not directly optimized and highlights the trade-offs involved in designing AI-assisted decision-making systems. Rather than producing a single universally fair solution, the proposed system helps organisers understand these trade-offs and supports more transparent decision-making.

Future work includes incorporating additional objectives such as experience balance and group-level fairness guarantees, exploring true multi-objective optimization methods that directly generate Pareto-optimal solutions, developing richer preference models, and evaluating the system using real participant data collected with appropriate consent. An interactive user interface could also improve the usability of the system for organisers without technical expertise.

**Reproducibility.** The full pipeline is regenerated by `python scripts/run_analysis.py`; individual steps are available as CLI commands (`generate`, `solve`, `evaluate`, `robustness`, `pareto`, `explain`). The dataset is generated with seed 42; method comparison uses seeds 42–46.

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